

Question

The elemental **X** reacts with oxygen at high temperatures to form a yellow solid **A**. Dissolving **A** in HClO_4 yields a perchlorate **B**. Upon dropwise addition of NaOH solution until neutral, a hexanuclear cation **C** with a +6 charge is obtained in the solution. This ion exhibits high symmetry, possessing four threefold axes, with all **X** atoms being tetrahedrally coordinated. Further addition of NaOH produces a white precipitate **D**. Continuing to add NaOH until alkaline and introducing Cl_2 yields a yellow precipitate **E**.

Element **Y** is a transition metal whose elemental form reacts with water under certain conditions to form a white precipitate **F**, which rapidly converts to a brown substance **G** containing approximately 52.35% **Y** by mass upon exposure to air. It is known that dissolving **F** in dilute sulfuric acid and then adding **E** results in a purple solution **H**.

Deduce the chemical formulas of each substance and determine which of the following statements is correct:

- A.** If the electron configuration of element **X** is written, its atomic core should be $[\text{Xe}]$.
- B.** The molecular point group of the cation **C** is consistent with that of methane.
- C.** The white precipitate **D** contains chlorine.
- D.** The mass fraction of element **X** in the yellow precipitate **E** is 80.25%.
- E.** During the process of **F** generating **H**, the oxidation state of the central element increases by 4.
- F.** None of the above options are correct.

Answer

Correct Answer: **F**

Detailed Explanation

First, **X** can react with oxygen when heated in air, indicating its relatively active chemical properties, which rules out the stable noble metals in air.

CHECKPOINT

1 PTS

X can react with oxygen when heated in air, so it is not a stable noble metal in air.

Its combination with oxygen should produce oxide **A**, which is stable in air and represents its primary oxidation state. **A** dissolves in perchloric acid, with only two possibilities: either no redox reaction occurs, or it is oxidized by perchloric acid to a higher oxidation state. However, perchloric acid is unlikely to perform oxidation reactions under normal conditions that are difficult for oxygen to achieve in dry heating, and perchloric acid also lacks the ability to alter redox potentials through coordination. Therefore, it is highly probable that the reaction from **A** to **B** is a non-redox reaction.

CHECKPOINT

1 PTS

The reaction from **A** to **B** is a non-redox reaction.

From **B** to **C**, a hydrolysis reaction occurs, forming a hexanuclear complex with four threefold axes. First, consider the O_h point group, i.e., a regular octahedral shape. Given the possible oxidation states of **X**, it cannot be +1 (as no hydroxide or oxide ions would form), but it could be +2 or +3. Oxidation states of +4 or higher would make it difficult to propose a reasonable structure with hydroxide or oxide ion counts that match the symmetry.

D is clearly the precipitate of the fully hydrolyzed hydrate, while the yellow precipitate **E** is more likely the product of a redox reaction with chlorine gas. Therefore, **E** should be a strong oxidizing agent.

CHECKPOINT

1 PTS

E should be a strong oxidizing agent.

Based on the subsequent text, **F** is oxidized under acidic conditions by **E** to produce a purple-red solution, which immediately suggests the permanganate ion MnO_4^- . Thus, it is hypothesized that **Y** should be manganese. Following this logic, **F** is manganese(II) hydroxide $\text{Mn}(\text{OH})_2$, and the condition "when exposed to air, it immediately transforms into a brown substance **G** containing **Y** with a mass percentage of approximately 52.35%" can be used to verify the hypothesis, calculating **G** as $\text{MnO}(\text{OH})_2$. The oxidation state of Mn^{2+} increases from +2 to +7 in MnO_4^- , a rise of 5, so option E is incorrect.

CHECKPOINT

1 PTS

F is $\text{Mn}(\text{OH})_2$.

CHECKPOINT

1 PTS

The chemical formula of **G** is $\text{MnO}(\text{OH})_2$.

CHECKPOINT

1 PTS

Mn^{2+} is oxidized to MnO_4^- , with the oxidation state increasing from +2 to +7, a rise of 5, so option E is incorrect.

There are few strong oxidizing agents capable of oxidizing Mn(II) to MnO_4^- under acidic conditions. With some thought, the most likely candidate is NaBiO_3 , as only bismuth can achieve the +3 oxidation state (Bi(III)) before being oxidized by chlorine gas, and it forms a characteristic sodium ion precipitate.

CHECKPOINT

2 PTS

Based on oxidizing ability and solubility, **E** is determined to be NaBiO_3 .

Calculations show that the mass percentage of **Bi** in **E** (NaBiO_3) is approximately 74.64%, while the given data in the question is 80.25%, indicating an error, so option D is incorrect. From this, the central element of the problem is deduced to be bismuth. Bismuth is in the sixth period with an electron core of $[\text{Xe}]$, not $[\text{Kr}]$, so option A is incorrect.

CHECKPOINT

1 PTS

The mass percentage of **Bi** in **E** (NaBiO_3) is approximately 74.64%, so option D is incorrect.

CHECKPOINT

1 PTS

The electron core of bismuth is $[\text{Xe}]$, so option A is incorrect.

Proceeding sequentially, **X** is oxidized in air to form the yellow oxide



(**A**), which then dissolves in perchloric acid in a non-redox reaction to form



(**B**).

CHECKPOINT

1 PTS

A is



, and **B** is



.

Next, using oxidation state analysis, the question provides the number and oxidation state of

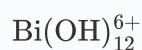


(+6), allowing the number of hydroxide ions to be easily determined as 12 based on charge balance. The possibility of oxygen atoms is also considered, but replacing hydroxide ions with oxide ions does not yield a structure with high symmetry.

CHECKPOINT

1 PTS

C is



, and replacing hydroxide ions with oxide ions does not yield a structure with high symmetry.

In reality, this forms an ion resembling a cube, where



occupies all face centers and



occupies all edge centers, consistent with O_h (regular octahedral) symmetry. Methane, however, has T_d (tetrahedral) symmetry, so they are inconsistent, and option B is incorrect.

CHECKPOINT

2 PTS

The point group of $\text{Bi}_6(\text{OH})_{12}^{6+}$ is O_h (regular octahedral), which is inconsistent with methane's T_d (tetrahedral), so option B is incorrect.

Throughout the process, perchlorate ions do not participate in redox reactions and do not produce chloride ions, so **D** does not contain chloride ions and is



, making option C incorrect.

CHECKPOINT

1 PTS

D does not contain chloride ions and is



, so option C is incorrect.

Thus, the following can be summarized:

X:



A:



B:



C:



D:



E:



F: $\text{Mn}(\text{OH})_2$

G: $\text{MnO}(\text{OH})_2$

H: NaMnO_4

Analysis of the options is as follows:

A: The electron configuration of Bi is $[\text{Xe}]4f^{14}5d^{10}6s^26p^3$, with an electron core of $[\text{Xe}]$, not $[\text{Kr}]$. Incorrect.

B: The point group of **C** ($\text{Bi}_6(\text{OH})_{12}^{6+}$) is O_h (regular octahedral), while methane is T_d (tetrahedral), so they are inconsistent. Incorrect.

C: **D** ($\text{Bi}(\text{OH})_3$) does not contain chlorine. Incorrect.

D: The mass percentage of Bi in **E** (NaBiO_3) is approximately 74.64%, while the given data is 80.25%, which may be erroneous. Incorrect.

E: Mn^{2+} is oxidized to MnO_4^- , with the oxidation state increasing from +2 to +7, a rise of 5, not 4. Incorrect.

Therefore, options A-E are all incorrect, and option F is correct.